Teaching a Small Model to Deliberate

Verifier-Anchored Reinforcement Learning over a Frozen LLM

2026

Cheap models, expensive reasoning, gamed rewards

- Small 1–7B models are fast and deployable, but lag on GSM8K and MATH
- Test-time search (best-of-N, self-consistency, MCTS) helps but is amnesiac: nothing is distilled back, every query pays again
- Dense *learned* reward models close the gap by opening a Goodhart attack surface
- Our answer: learn a small policy; the teacher is search; the reward is a non-LLM verifier

Thesis

1. Freeze the base LLM; learn small heads π_θ, Q_ψ over a structured state
2. Cast LLM deliberation on verifiable tasks as approximate Q-learning over an induced context MDP
3. Reward is verifier-anchored, not a learned critic
4. MCTS is a sample-efficient *policy-improvement operator* supplying stronger targets, not the central algorithm

The induced context MDP: structured state + seven verbs

- **State** s: problem, Facts, open/closed Subgoals, Hypotheses, verifier log, answer, depth
  - load-bearing split: **Fact** (proven) vs **Hypothesis** (tentative)
- **Actions** A ($|A|=7$): `DECOMPOSE`, `COMPUTE_SUBEXPR`, `FORM_HYPOTHESIS`, `CHECK_CASE`, `INVOKE_VERIFIER`, `BACKTRACK`, `COMMIT_ANSWER`
  - only the verb is categorical; the frozen LLM instantiates content (mostly prompt templates); a state mask gates illegal verbs
- **Transition**: frozen LLM + template *samples* the next state (non-enumerable)
- **Reward**: ground-truth terminal $+1/-1$ from the verifier, plus a small capped intermediate shaping signal

A concrete trace (GSM8K)

P: "A store sold 7 boxes with 6 pencils each. How many pencils?"

1. `FORM_HYPOTHESIS`: `hyp_1` = "the total is 42" (tentative)
2. `COMPUTE_SUBEXPR` "7×6" $\rightarrow$ 42 (derived Fact)
3. `INVOKE_VERIFIER` on `hyp_1` (SymPy) $\rightarrow$ confirmed; promoted to Fact ($+0.05$, capped)
4. `COMMIT_ANSWER` "42" $\rightarrow$ terminal verifier $\rightarrow$ reward $+1$

Architecture: frozen LLM, tiny heads, offline by design

- One frozen LLM is *both* encoder and stochastic generator (one forward); embedding = hidden state at `[STATE_END]`
- Trainable: a small adapter + multi-head $Q_\psi(s, a)$ over the 7 verbs + a policy-prior head π_θ — under 5M params
- Approximate Q-learning: replay buffer, target net, prioritized by TD error
- Q-learning is off-policy $\Rightarrow$ learns from *any* trajectories: its own, other models', logs, human demos

MCTS as a policy-improvement operator, and the loop

PUCT (learned policy as prior, Q_ψ as leaf value):

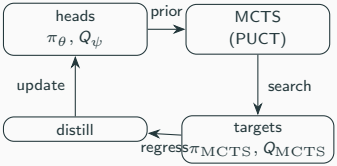
$$\arg\max_a \ \widehat{Q}(s,a) + c_{\text{puct}}\,\pi_\theta(a \mid s)\frac{\sqrt{N(s)}}{1+N(s,a)}$$

Chance-node mean is an honest Bellman estimator:

$$\tfrac{W(s,a)}{N(s,a)} \approx \textstyle\sum_{s'} P(s' \mid s,a)[R + \gamma V(s')]$$

Distill targets: π_{MCTS} (visits), Q_{MCTS} (chance-Q);

$$\mathcal{L}_{\text{total}} = \left(Q_\psi - Q_{\text{MCTS}}\right)^2 + \lambda\,\mathcal{L}_\pi$$



System 1 heads, System 2 search; training shifts competence into System 1.

Reward hacking: a verifier is necessary, not sufficient

- Goodhart: optimize a mis-specified reward, get behavior that scores well and reasons badly
- A non-LLM verifier removes the *learned*-reward attack surface, but a poorly specified verifier reward is still gameable
- We red-teamed our own intermediate shaping reward and found a farm: verifying content-free or self-authored claims for free reward

Fix by construction, not by patch

Tie the rewarded check to the target claim; reward only the *first* success per target; hard-cap cumulative intermediate reward at $\frac{1}{5}$ of the terminal $\Rightarrow$ the ground-truth terminal verifier always dominates, so farming is *bounded*, not just discouraged.

Inference and deployment: one compute knob, always-on learning

$B = 0$ (fast)

masked arg max over Q_ψ; one forward/decision; constant latency

$B > 0$ (search)

PUCT MCTS for B sims; sample the visit-count policy

- Same trained heads; one dial traces a compute-vs-accuracy frontier, no retraining
- Off-policy + verifier-anchored $\Rightarrow$ keep learning in deployment from any verifier-style signal, incl. user thumbs up/down
- Per-user adapters absorb idiosyncratic feedback; the verifier stays the anchor that bounds drift

Positioning and defensible novelty

- **AlphaZero / expert iteration**: we invert it — agent and Q_ψ central, search optional
- **Verifier-anchored reasoning** (PRMs, AlphaProof, DeepSeek-R1): verifier-vs-aesthetic asymmetry is load-bearing; `INVOKE_VERIFIER` is a callable verb
- **VerifierQ**: verifier-as-*action* not verifier-as-Q; online distilled targets; categorical multi-head Q
- **vs STaR / ReST-EM / Math-Shepherd**: per-step Q_{MCTS} as a genuine Bellman target, not trajectory-level or a process-reward model

How we will evaluate (the plan, not results)

- GSM8K and MATH; baseline grid fixed in advance: base-greedy, best-of-N, self-consistency, STaR/ReST-EM, AlphaZero-for-reasoning, PRM-guided
- Pre-registered hypotheses and the headline frontier figure committed *before* training; ≥ 3 seeds, ≥ 8 replicates, mean $\pm$ std
- Holm–Bonferroni across the FLOP $\times$ benchmark grid; explicit compute-leak threat model

Honest frontier accounting

x: total inference FLOPs (charge *every* forward inside the B simulations, not just the root); y: accuracy; FLOP-matched curves. Pilot: generation dominates prefill $\approx$ 79:1 per decision.

Vision

A small, fast model that learns to deliberate over structured states,
anchored by verifiable (non-learned) rewards.

Offline today, online-continual in deployment, one test-time compute knob.

MCTS earns its keep only as the operator that manufactures stronger targets.